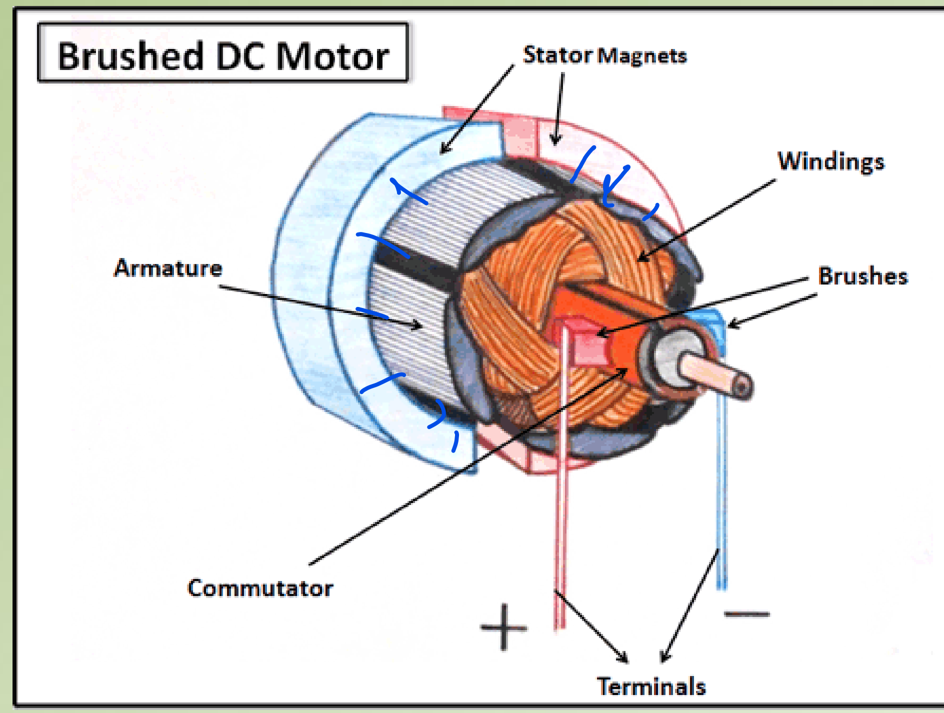
Electromechanical Characteristics of Brushed DC Motors

Permanent Magnet Brushed DC Motors

- One of the most common types of actuators in electromechanical systems
- Characteristic constants for a DC motor:
  - torque constant and speed constant
- Speed, torque, power, and efficiency curves

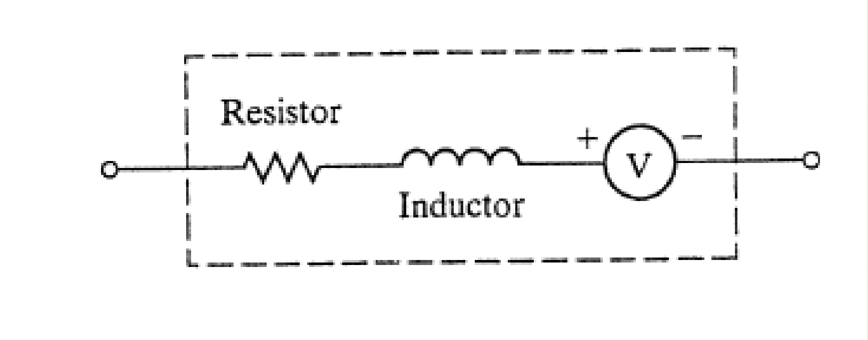
Brushed DC Permanent Magnet Motors

- Sub-fractional horsepower devices
- Low cost, and easy to use
- Operate using DC voltages/currents



ELECTRICAL MODEL

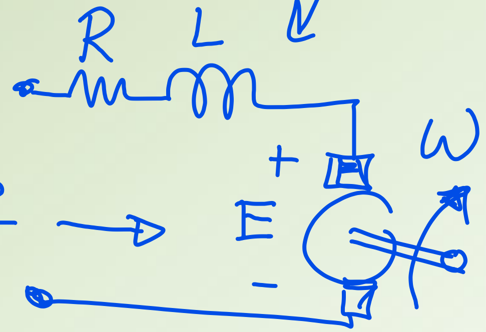
- Modeled by a resistor, an inductor, and a source of electromotive force (EMF)
- EMF is also be called back-emf or counter-emf
- Where do the R & L come from?



Back EMF (Faraday's Law):

The motor acts as a generator at the same time that it acts as a motor.

back - emf →



Torque and speed equations

- *Ke* depends on the construction, geometry, and materials properties of the motor.

$$E = K_e \omega$$

where: E = back-EMF [V]

K_e = speed constant $\left[\frac{V}{rad/s}\right]$

ω = rotational speed [rad/s]

The torque generated by the motor consists of two terms:

- Frictional torque
- Usable torque

$$T_M = T_L + T_f \qquad [Nm]$$

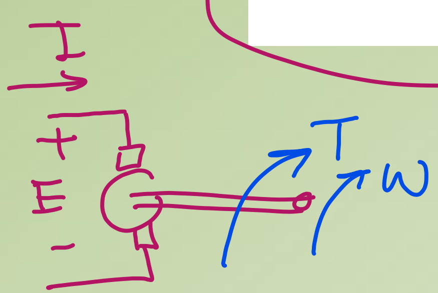
➔ Neglecting friction, the electrical power is equal to the mechanical power and we have

$$T = K_T I \qquad [Nm]$$

where: T = torque [Nm]

K_T = torque constant [Nm/A]

I = current [A]



electric power

$\rightsquigarrow$ balance power $\rightarrow EI \approx T\omega$

$$K_e \cancel{\omega} \cancel{I} = K_T \cancel{I} \cancel{\omega} \rightarrow K_e = K_T$$

… Torque and speed constants

When compatible units are used ➔ $K_T = K_e$

When the constants are expressed in inconsistent units (more common in practice), a conversion factor must be applied to convert between them.

Two of the most commonly needed conversions are as follows:

$$K_T \text{ [in}\cdot\text{oz/A]} = 1.3542 \times K_e \text{ [V/krpm]}$$

$$K_T \text{ [Nm/A]} = 9.5493 \times 10^{-3} \times K_e \text{ [V/krpm]}$$

oz stands for ounce-force:

1/16 of a pound-force, or 0.2780139 newton

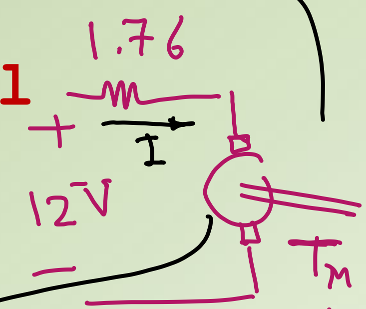
Example: A motor with *KT* = 5.89 *in.oz/A* and a coil resistance of 1.76Ω is driven with a supply voltage of 12 V.

If the motor's friction torque is 1.2 *in.oz*, what is the maximum torque available for driving a load?

$T_m = K_T I$

Maximum torque: When speed = 0 (stall torque)

when $E = 0$



$E = K_e \omega$

max = ?

➔ **I=12/1.76=6.82 *A***

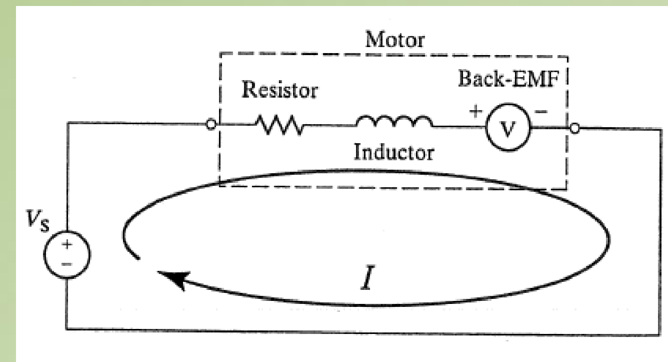
$$T_M = T_L + T_f = K_T I$$

$$T_L = K_T I - T_f$$

➔ **TL= KT*I-Tf =5.89*6.82 - 1.2 = 39 *in.oz***

Speed, power, and efficiency characteristics vs torque

Using DC motor equations in the steady state and solving for speed in terms of torque, we get the speed-torque characteristic shown in the formula/figure below

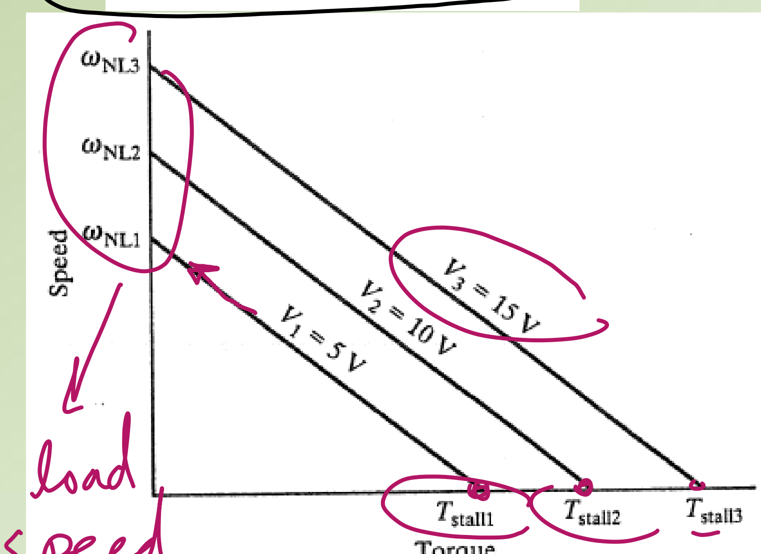


$$\omega = \frac{V}{K_e} - \frac{R}{K_T K_e} T$$

$$\omega \uparrow \longrightarrow I \downarrow \longrightarrow T \downarrow$$

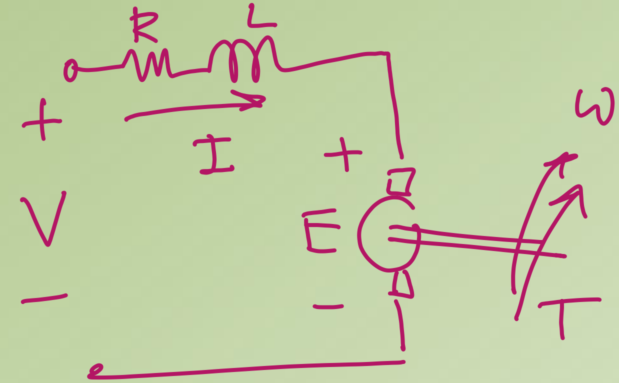
- Max speed (no load speed) corresponds to zero current/torque
- When *speed=0, I and T are maximum* ("stall")

no load speed



$E = K_e \omega$ (1)

$T = K_T I$ (2)



Neglect effect of L.

KVL $V = RI + E$ (3) $\Longrightarrow I = \dfrac{V - E}{R}$ (4)

Sub. (4) in (2) $\longrightarrow$ $T = K_T \dfrac{V - E}{R}$ (5)

Sub (1) in (5) : $T = K_T \dfrac{V - K_e \omega}{R}$

Solving for ω $\Longrightarrow$ $\omega = \dfrac{1}{K_e} V - \dfrac{R}{K_T K_e} T$

Example: **A servo application requires at least 225*mNm* of torque at 2000*rpm*. Is a motor with a no-load speed of 9550rpm and coil resistance of 2.32Ω, powered by a 24V battery, sufficient for this application?**

$$\omega = \frac{\overset{24}{V}}{K_e} - \frac{R}{K_T K_e}\overset{0}{T}$$

when T=0

- No load speed = *9550rpm* ➔ Set *T=0, V=24* ➔ *Ke= 24/9550* V/rpm = *(24/9550)*(60/(2π))=***0.024Nm/A**

- At 2000 rpm: $\underbrace{2000*2*\pi/60}_{\omega} = \underbrace{24}_{V}/\underbrace{0.024}_{Ke} - \underbrace{2.32}_{R} * T/0.024^2$
  → T= 0.196 Nm

- Can the motor do the job?

< 0.225 Nm

Power-torque characteristics

$$P = \underline{T}\omega = (T_f + T_L)\,\omega \qquad [\mathrm{W}]$$

Friction

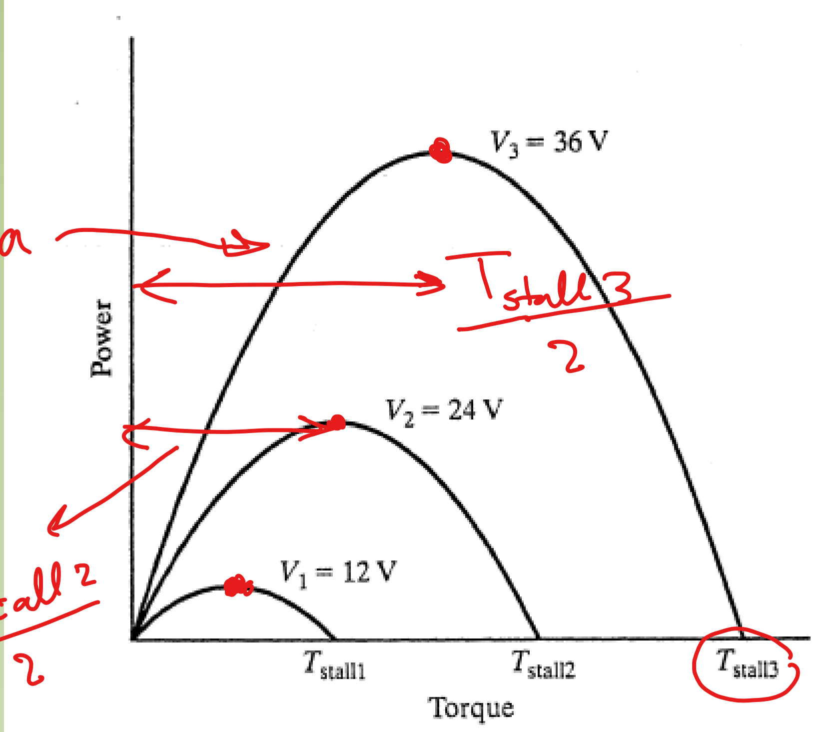
$$P = T\left(\frac{V}{K_e} - \frac{R}{K_T K_e}T\right)$$

→ parabola

Differentiating *w.r.t.* T yields the maximum power point (occurring at Tstall/2) with:

$$P_{max} = \left(\frac{K_T}{4K_eR}\right)\cdot V^2$$

$\frac{T_{stall\,2}}{2}$

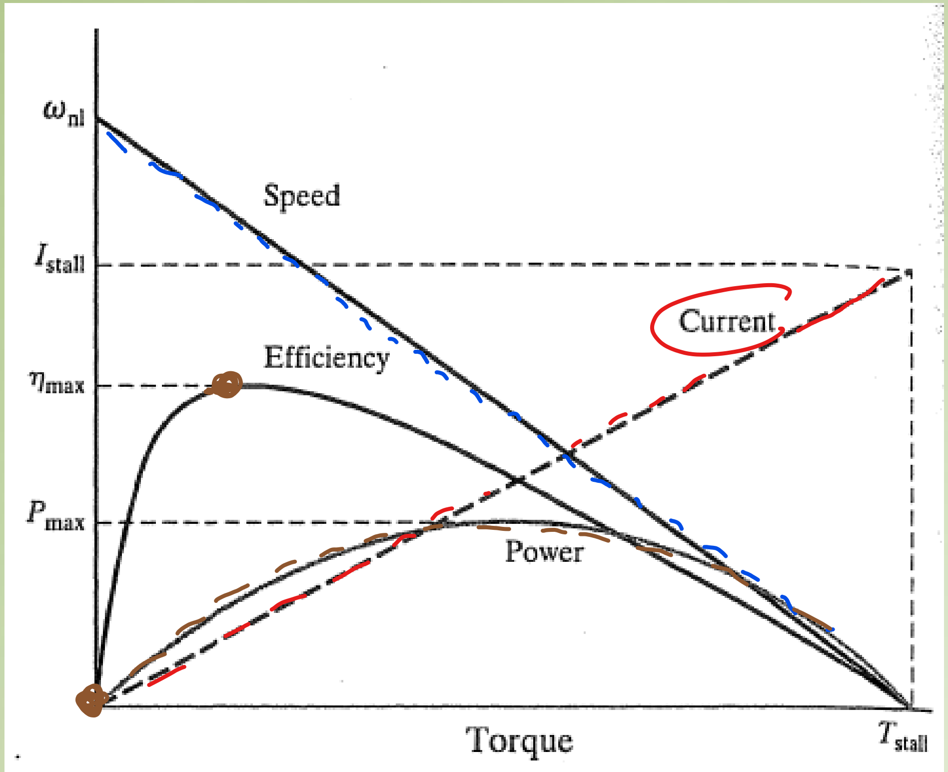


Important result: *The mechanical power output changes as the square of the applied voltage.*

➔ Changes in voltage have a substantial impact on a motor's power output.

DC MOTOR EFFICIENCY

- At high torques (high currents), Ohmic losses are high and therefore efficiency is low.
- At very low torques (high speeds), small useful mechanical power is produced (mostly used to overcome friction) which also leads to low efficiency.
- Efficiency is maximum in between

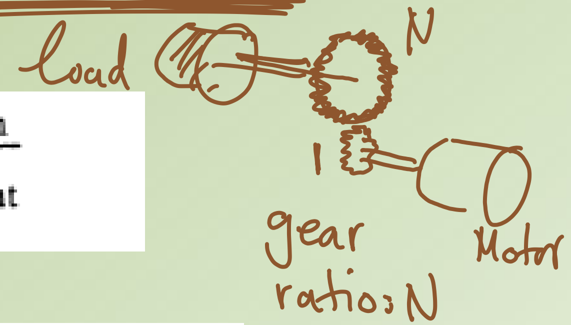


In general, ***maximum efficiency*** operation occurs at relatively ***high speed*** and ***low*** *values of* ***torque*** (current) leading to minimal Ohmic losses.

Gearheads

- The ***most efficient operating currents/torques*** lie ***above no-load current, and below the condition for maximum power*** (page before)
- Many applications require substantially lower speeds and higher torques

➔ gearheads ➔ $$\text{Gear ratio} = N = \frac{\omega_{\text{in}}}{\omega_{\text{out}}}$$



Gearhead efficiency should be considered (e.g., 50% for low-end, >90% for smaller reductions (e.g., 4:1)

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{T_{\text{out}}\omega_{\text{out}}}{T_{\text{in}}\omega_{\text{in}}}$$

$$T_{\text{out}} = \eta T_{\text{in}} N$$

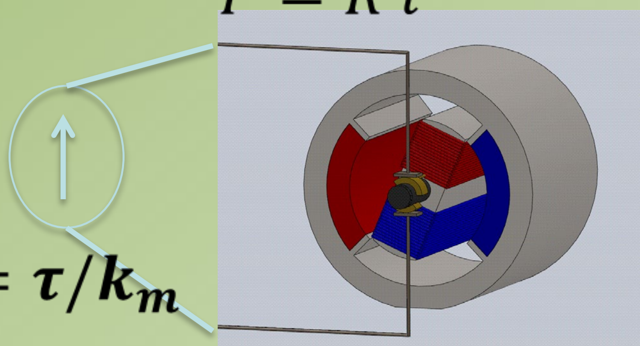
torque transmission efficiency

Thermal Model of a DC Machine

- Heat dissipation due to resistive losses in the motor coils (resistance R)

➔ *Insulation coil temperature < 80 ° C*

$P = R\,i^2$

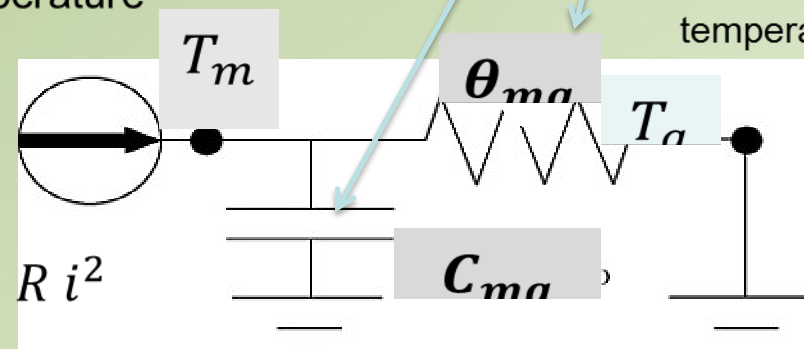


$\boldsymbol{i = \tau / k_m}$

T_m
Motor winding temperature

Thermal resistance and capacitance

T_a
Ambient temperature

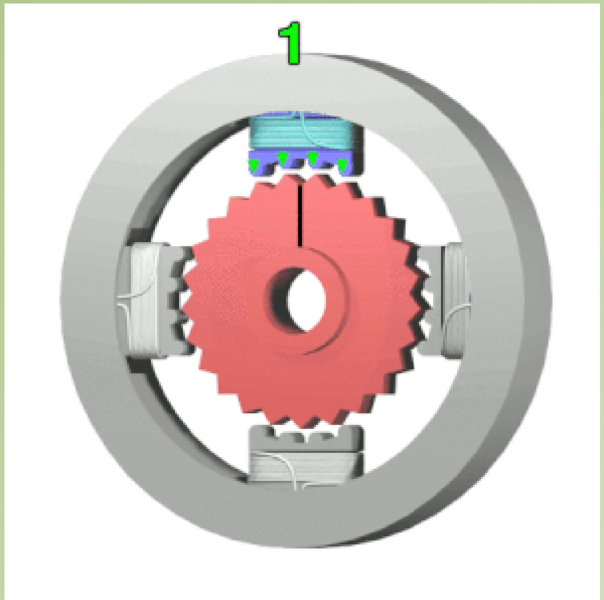


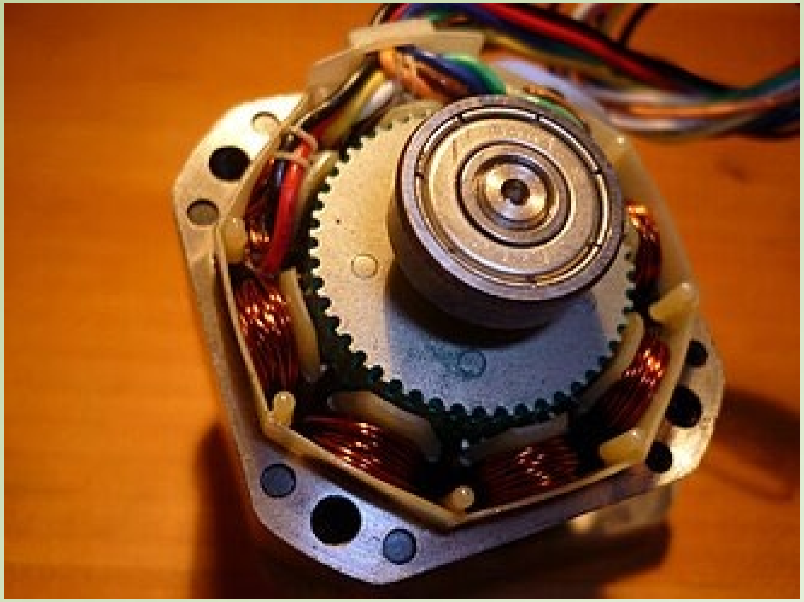
$P = R\,i^2$

$$T = T_m - T_a \Rightarrow P = C_{ma}\frac{dT}{dt} + \frac{T}{\theta_{ma}} \;\Rightarrow\; \frac{T(s)}{P(s)} = \frac{\theta_{ma}}{1 + sC_{ma}\theta_{ma}} \;\Rightarrow\; \frac{T(s)}{I^2(s)} = \frac{R\,\theta_{ma}}{1 + sC_{ma}\theta_{ma}}$$

Stepper motors:
See animation in Wikipedia:
https://en.wikipedia.org/wiki/Stepper_motor#/media/File:StepperMotor.gif

Electromagnets are energized to move the motor





- A stepper motor is synchronized using direct current -its rotation is comprised of a large number of steps.
- The position of the motor can be controlled without having to resort to closed-loop control (feedback)- need a lot of *steps* for better accuracy
- Stepper gear motors reach the maximum torque at low speed, which makes them particularly suitable in high-precision applications that require high holding torque such as robotics, 3d printers, and all devices where position accuracy is critical.

Stepper motors vs DC Motors:

- Are heavier
- Use more power
- Have less torque than DC motors.
- Cannot be operated at high speeds, i.e., the faster they run, the less torque they have

- **Generally speaking:**
    - Servo motors are more efficient than stepper motors (80-90% vs lower for steppers)
    - Servo motors are best for high speed, high torque applications while stepper motors are better suited for lower acceleration, high holding torque applications.

CHARACTERISTIC	DC GEAR MOTORS	STEPPER GEAR MOTORS
Loop	closed loop	open loop
Efficiency	high efficiency	low efficiency at maximum load
Control	not required	microcontroller required
Speed	wide range of speeds	below 2000 RPM
Brushes	brushed	brushless
Motion	continuous	incremental

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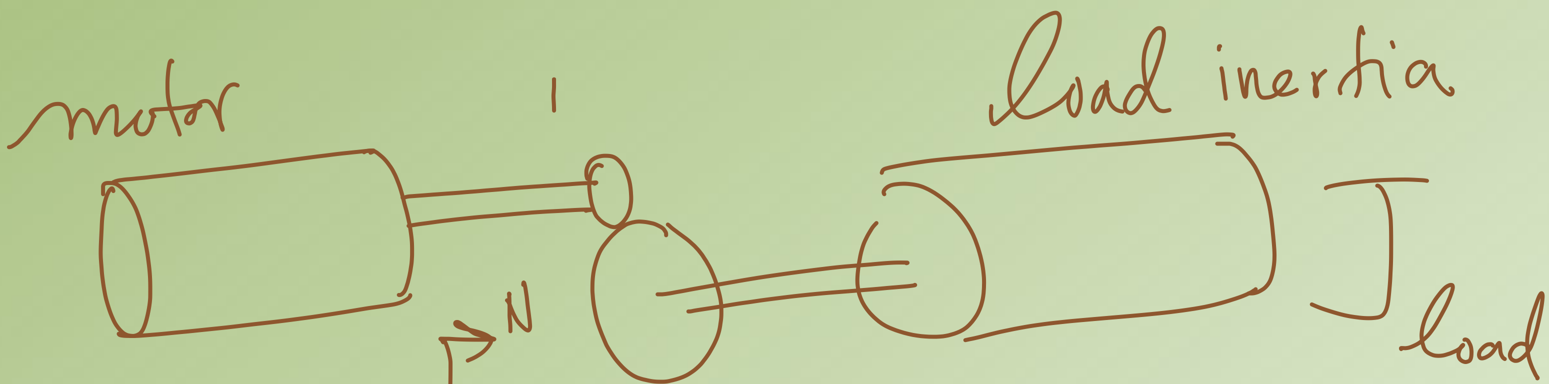

Total inertia experienced by motor

$$J_{total} = J_{motor} + \frac{J_{load}}{N^2}$$